American International University-Bangladesh (AIUB)
Department of Computer Science

Faculty of Science & Technology (FST)

PROJECT TITLE

AIUB Canteen Management System

Semester: Fall 25-26

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1.1 Background to the Problem

The university canteen is one of the busiest service points on campus, serving hundreds of students, faculty members, and staff every day. It plays a crucial role in meeting the daily dining needs of the campus community. Despite this importance, many university canteens still rely on outdated and inefficient manual methods for recording sales, managing inventory, and issuing bills. These traditional processes often lead to unnecessary complexities and operational inefficiencies. Over time, issues such as billing mistakes, poor stock management, and lack of transparent financial reporting become more frequent. As the student population grows, these problems become even more noticeable.

At the American International University-Bangladesh (AIUB), the situation mirrors these challenges. Most of the sales and inventory tasks are still handled manually, causing various operational delays. Since the limited food outlets are not automated, even simple orders like a sandwich take longer to process, resulting in long queues during busy hours. Manual pricing increases the likelihood of errors, leading to customer dissatisfaction and financial discrepancies. The absence of digital record-keeping makes it difficult for the authorities to evaluate canteen performance. Without real-time access to sales data and stock levels, staff cannot quickly identify shortages or remove items that are not selling. Additionally, the lack of a secure role-based access system poses security risks, as staff may perform tasks without proper authorization.

These inefficiencies slow down daily operations significantly. Canteen staff struggle to manage long lines, and students, teachers, and employees end up wasting valuable time waiting for service. Customers become frustrated with slow service, billing mistakes, and inconsistent food availability. Relying on paper-based documentation also puts pressure on workers, making it difficult to maintain accuracy during peak hours. As AIUB continues to expand, the demand on the canteen increases, and these problems will only worsen unless modern solutions are adopted.

To address these challenges, implementing a digital canteen management system has become essential. A secure, role-based, user-friendly desktop application can replace the current manual operations with a more efficient and reliable process. Such a system would allow Admins and Staff to log in with appropriate access levels, ensuring security and accountability. It would centralize all information, enabling real-time updates of inventory, sales records, and transaction histories. Automated billing, sales reports, and product management would minimize errors and streamline operations. Digital monitoring would also provide administrators with valuable insights to make informed decisions, improve inventory planning, and enhance service quality.

A well-designed system would not only simplify operations but also improve accuracy, reduce workload, and elevate the overall service experience for the AIUB community. By eliminating the limitations of manual processes, the canteen could operate more efficiently, serve faster, and provide a more satisfying experience for students, staff, and faculty members alike.

1.2 Selection of Process Model

The Waterfall model is chosen because the project requirements are well-defined and stable, making a sequential and structured approach suitable for cost estimation and timely delivery.

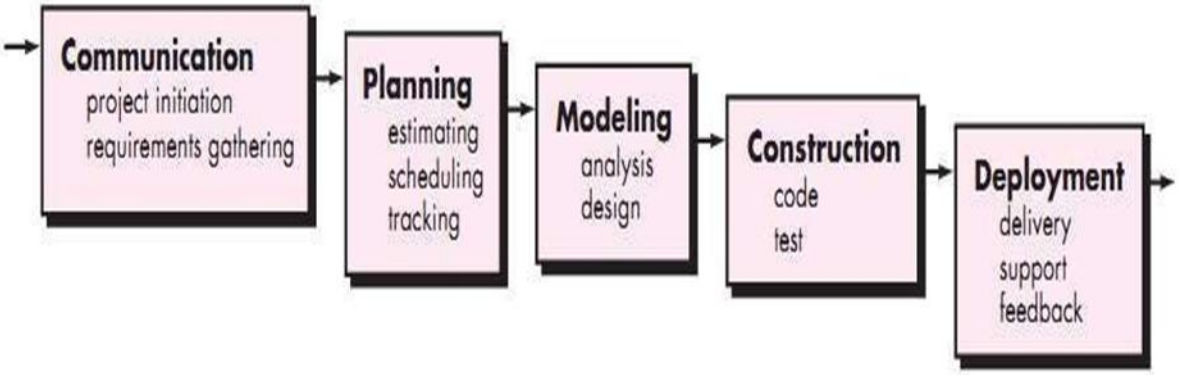

Fig 1: Diagram of Waterfall Model

Reason for Using the Waterfall Model

The Waterfall model is adopted here as the development model to be used for the AIUB Canteen Management System because in this project everything is clearly defined and described in advance. Here in the current assignment things like secure login, inventory tracking, sales processing, reporting and interaction with customers are already tabulated in SRS. As requirements will not be adjusted often in the duration of the project the consecutive nature of the Waterfall model allows a solid and secure development. And each phase: Communication, Planning, Modeling, Production and deployment can thus be carried out in succession ensuring a lined-up sequence where one process follows the next from idea to completion.

One of the best advantages of the Waterfall model here is that the cost and time estimation is improved. As each stage has clear start and finish boundaries it is more feasible to estimate the resources, deadline, and deliverable. This is particularly useful in an academic project, wherein a team usually has limited time, manpower, and funds. The model also emphasizes proper documentation at every stage which ensures the system design, testing methodology, and implementation details are documented. In the field of documentation not only does team coordination enhance but the system also becomes stabilized and easily modifiable in the long term. Another reason why the Waterfall model is favored since there are less confusion and risk for juniors or new developers. The team just focuses on one stage and another without the concern of changes in constant demand or their tasks overlapping. For instance, when achieving the requirements and design stage, developers are initially aware with the system that minimizes chances for errors in code. Testing becomes more efficient as it tests after complete implementation of the system unlike on partial builds. This structured approach increases the probability of building a robust and reliable canteen management solution for AIUB.

The Waterfall methodology suits both the educational character of the project as well as the documented requirements. It gives a very strict and manageable structure; it makes for good analysis of the cost involved, and it allows phase to be effectively completed before starting on the next. The clarity, risk, as well as the effectiveness of the AIUB Canteen Management System under constraints can be achieved by the development team.

Analysis of the Project Environment

The AIUB Canteen Management System environment for the project is quite stable as requirements are clearly described and documented in the SRS. Key features such as RBAC, inventory control, sales processing, reporting and feedback receiving are well defined and are unlikely to change during development. The latter is due to the fact that the existing process will be replaced by a new system and there are no real additional requirements from canteen operations not covered in the application. But, eventually, you may need additional functionalities like PSD integration for online ordering, support for mobile apps or a service that integrates with digital payment gateways as well. These possible changes indicate that, although the present conditions are stable, future releases might imply modified requirements based on user response and institutional demands.

Team Size, Communication, and Feasibility

Waterfall model are suitable for our small development team, and it takes an ordered way which each member can concentrate at a phase during its period (Communication, Planning, Modeling, Construction and deployment). This makes things less complicated and less likely for misinterpretation, because the work here is concrete and written down before you move to the next stage. By breaking down the work in the SRS and detailed design document, all your team members have access to a consistent reference point that reflects everything they ever need to know about client deliverables. The solution is workable to accomplish the goals of AIUB Canteen Management System. Wha Follow Waterfall, the team guarantees that the systems' most important goals- scanning sales process, managing inventory, taking orders and contributing to a better customer experience, are delivered on time. 2, 3) The successive process model regulates that all the business requirements are systematically processed and finally achieves a safe stable system for the university canteen's daily operation.

Flexibility of the Model

The Waterfall model is typically not very agile when it comes to changes to scope, technology, or user needs. When any phase (requirements or design) has been completed it is hard and expensive to go back to a step, simply because the entire process depends on one part of the sequence causing another. The requirements for this project are well defined and will not be subject to change, so this limitation does not allow too much short-term risk. But when new needs arise in future, for example mobile integration into app or ability to support digital payment system or even a feature to order online, the Waterfall cannot accommodate them on very easily terms without re-starting whole development scenario. For such cases, more flexible models like Agile Scrum would be suitable as it permits incremental change and ongoing user feedback.

Target Users and Benefits

The target group of users for the proposed solution includes Admins, Staff, and Customers (students, teachers, and employees) of the AIUB canteen.

- Admins benefit, with full control over the system, enabling them to manage inventory, staff accounts, sales reports and viewing feedback. With them, they can monitor transactions in real time, cut down on errors, and maintain transparency of financial records.
- Staff have an easier, quicker sale process with the cart POS system that supports quick order editing, mass sales orders, and on-the-spot invoicing. This relieves workload and the human barrier and increases service speed.
- Customers will be able to experience a seamless user journey from coming in; clear pricing – printed/digital receipts of order amendment before billing, and refunds/feedback. Plus, they benefit from exclusive promotions and loyalty programs, enhancing their experience overall. By managing the needs of each user group, the system ensures a more efficient, clear, and user-friendly canteen operation, improving both staff's productivity and customer experience.

Risk and Uncertainty Management

The Waterfall model mainly manages risks and ambiguity through its linear and systematic phases. The risks of misunderstanding the system's requirements are minimized at the requirement phase by developing a detailed and well-documented SRS that must explicitly specify what Admin, Staff or Customer functions entail. To manage the overall risk in DBMS design, technical risks are lowered by system architecture and database design is done beforehand, so all modules must interconnect well. During the implementation phase, risks associated with a bug in the code are mitigated by strict adherence to design guidelines. As testing occurs after completing implementation, the model guarantees all hazards of dysfunctional behaviors, data inconsistency or security breaches are found and resolved before deployment. Lastly, at the deployment and maintenance end of things operational failure risks are mitigated through appropriate documentation training and backup plans. Even if the Waterfall model does not contain appropriate mechanism for accommodating changes in requirements, its strong focus on planning, documentation and sequential development contribute to reducing risks and uncertainties in projects such as this one, where the requirements are already clear and stable.

Justification of the Chosen Model

Waterfall model is more appropriate in this project than others as the requirements of AIUB Canteen Management System are frozen not dynamic and will not be modified during the time The requirement will drag throughout the development process Whereas Agile or Spiral Models are designed for Projects that have Changing requirements. The Waterfall Model provides a structured and disciplined approach sequence of phases in which each phase is completed before next. This system is clear, not confusing, and highly manageable for a small academic team on limited resources and time. Waterfall also permits more accurate cost estimation and scheduling

(when compared to Incremental or Agile Scrum) because the full scope of work is known at the beginning. The attention to documentation at all levels means that the system design and implementation and tests are well-set-out, which is important for academic review as well as future maintenance. Although models such as Scrum or XP that incorporate more flexibility exist, they involve regular meetings, dynamic priorities and continuous stakeholder involvement, none of which is likely to be possible in this academic environment. Thus, the Waterfall model is most suitable for the environment of the project, since it delivers a reliable solution by following clearly structured and time-efficient direction to address canteen's operational requirements.

2. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS)

2.1 Scopes and Features

Users:

1. Admin
2. Staff
3. Customers

Scopes:

1. Profile Management

1. Secure login for Admin and Staff.
2. Update personal profile (Admin & Staff; User ID and role cannot be changed).
3. View login history of all users (Admin only).
4. Manage staff accounts: create, update, delete (Admin only).
5. Log out from system (Admin & Staff).
6. Remote logout of all staff (Admin).

2. Inventory Management

1. Add, update, search, and delete inventory items.
2. Set product discounts or combo offers.
3. Search products by name or category.
4. Flag low-stock or missing items (Staff).
5. Report damaged or wrong-priced items.
6. Print inventory stock summaries.

3. Sales & POS (Point of Sale)

1. Process customer orders via cart interface.
2. Edit cart items before finalizing sale.
3. Generate digital or printed invoices.
4. Reprint previous invoices.
5. Apply discounts (if permitted by Admin).
6. View prices, combos, and menu items.
7. Suspend/hold orders and resume later.
8. Calculate totals with taxes, discounts, and combos.
9. Customer can modify order before billing.
10. Customer can cancel order or request refund before completion.

4. Customer Interaction

1. Place orders at counter.
2. Receive printed or digital invoice.
3. Request invoice via SMS or Email.
4. View prices and pay.
5. Check estimated wait time.
6. Participate in loyalty/feedback discounts.
7. Report missing items or suggest items.
8. Submit feedback or complaints.

5. Reporting & Analytics

1. View sales history (Staff: personal only).
2. View full sales reports (Admin: daily, weekly, monthly).
3. Generate closing reports.
4. Search transactions by item, staff, or date.
5. View today's sales summary (Staff).
6. Export sales and inventory data to Excel/PDF.

6. System Configuration

1. Configure system preferences (tax, currency, business hours).
2. Assign permissions to staff (Admin).
3. Maintain data accuracy through real-time updates.

2.2 User Story Table

As A/An	I want to	So that	Acceptance Criteria
Admin	Login securely using admin credentials	I can access the admin dashboard safely	Admin can log in only with valid credentials; incorrect credentials are rejected with error messages; successful login redirects to admin dashboard.
	Manage staff accounts	I can create, update or delete user access	Admin can create new staff accounts with unique credentials; update existing staff info; delete accounts; reflect changes immediately.
	Add, update, search, and delete inventory	Product data remains accurate and up-todate	Inventory items can be added, edited, searched by name/category, and deleted; all changes are saved and visible in real time.
	View sales reports	I can monitor daily, weekly, and monthly performance	Admin can generate reports by date range (daily/weekly/monthly); reports display correct totals and itemized sales.
	Generate and review invoices	All transactions are properly documented	Every transaction generates an invoice with date, items, totals, and discounts; invoices can be reviewed later.
	Update personal profile	My contact info stays current (User ID C role unchangeable)	Admin can edit personal details except User ID and role; updates are saved successfully.
	Export sales/inventory to Excel/PDF	I can share or archive data easily	Admin can export reports in Excel or PDF format; files open correctly and match system data.

	Set product or combo discounts	I can run promotions and combos effectively	Admin can create/edit discounts or combos; discounts apply correctly during transactions.
	View login history of users	I can ensure security by checking access logs	Admin can view list of staff login attempts with date/time; failed logins are recorded.
	Configure application preferences	The system works as per canteen policy (tax, currency, hours)	Admin can configure tax, currency, and hours; changes apply across system.
	Generate closing reports	I can reconcile and close daily accounts	Closing reports include sales totals, discounts, taxes; generated at end of day.
	Search transactions	I can find specific records by item, user, or date	Admin can filter/search transactions by keyword, user, or date range; correct results displayed.
	Print inventory stock summaries	I have physical records of stock	Admin can print stock report; printed copy matches system data.
	Log out all staff remotely	I can handle urgent reset/logout situations	Admin can log out all staff with one action; staff sessions end immediately.
	Assign limited permissions to staff	I can control what staff can or cannot access	Admin can assign role-based permissions; restricted staff cannot access blocked modules.
	View all feedback and complaints	I can review and improve service quality	Admin can access customer feedback list; feedback is stored with date/time.
Staff	Login securely with staff credentials	I can access the POS system	Staff can log in only with valid credentials; invalid attempts are rejected.
	Process customer orders via cart	I can complete transactions quickly	Staff can add/remove items in cart; system calculates totals correctly.

	Edit cart items before sale	Mistakes can be corrected before payment	Staff can modify or remove items before finalizing transaction.
	Generate digital invoice calculate totals	Customers get a proper bill	Each sale produces invoice with totals, discounts, taxes; invoice can be printed/digital.
	View personal sales history	I can track my performance (7, 30 days, etc.)	Staff can view their sales summary for 7 or 30 days; data is accurate.
	Update personal profile	My contact info remains accurate	Staff can update personal details except User ID/role.
	Search products by name/category	I can serve customers faster	Staff can search products quickly by name/category; results show instantly.
	Apply discounts (if allowed by Admin)	I can provide offers when authorized	Staff can only apply discounts if permission is enabled; discount applies correctly in invoice.
	Reprint previous invoices	Customers can get duplicate receipts	staff can search past invoices and reprint exact copies.
	Flag low stock or missing items	Admin can take inventory action	Staff can mark items as low stock/missing; notifications sent to Admin.
	View today's sales summary	I can monitor daily earnings	Staff can generate daily sales summary; totals match actual transactions.
	Suspend/hold orders temporarily	I can pause complex or incomplete orders	Staff can hold an order, save it, and resume later without data loss.
	Report damaged or wrong- priced items	Product info and quality stays correct	Staff can report errors in pricing or damaged items; report sent to Admin.

Customer	Place orders at the counter	I can get food and drinks I want	Customer can place order easily; staff confirms and processes it.
	Receive printed or digital invoices	I have proof of payment	Every completed order generates invoice; invoice can be printed or sent digitally.
	View prices and pay	I know how much to pay	Prices are visible before checkout; final bill is clear and accurate.
	Request invoice via SMS or Email	I can get receipt copy later if needed	Customers can request invoice via SMS/email; copy is delivered correctly.

	Browse menu or combo offers	I can choose my preferred items easily	Menu and combo offers are displayed clearly; selections update cart.
	Modify order before final billing	I can change my decision in time	Customer can request changes before final billing; cart updates accordingly.
	Check estimated wait time	I can plan accordingly	System displays estimated preparation/wait time; updates as needed.
	Participate in loyalty/feedback discounts	I get rewards or benefits	Loyalty discounts apply automatically if criteria met; benefits are visible on bill.
	Suggest or report missing items	The menu can be improved	Customers can submit suggestions or missing items; feedback is stored.
	Cancel or request refund	I feel secure if my order is delayed or wrong	Orders can be canceled or refunded before completion; record stored for Admin.

	Provide feedback or complaints	Service quality can improve	Customers can submit feedback after transaction; stored for Admin review.

2.3 Requirements Traceability Matrix

2.3.1 Functional Requirements

2.3.1.1. Admin Requirements

1.0 User home page-All features availability	
1.1	**Login**
1.1.1	Taking username
1.1.2	Taking password
1.1.3	Providing forgot password option
1.1.3.1	Submitting e-mail address and receiving reset link or code
1.1.4	Submitting the login form
1.2	**Dashboard**
1.2.1	Side menu
1.2.1.1	Dashboard button [ref 1.2]
1.2.1.2	View profile button
1.2.1.2.1	Show name
1.2.1.2.2	Show e-mail
1.2.1.2.3	Show phone number
1.2.1.2.4	Show role
1.2.1.2.5	Show profile picture

1.2.1.2.6	Edit profile button
1.2.1.2.6.1	Show existing info and allow editing (name, email, phone)

1.2.1.2.6.2	Submit updated info
1.2.1.2.6.3	Return to profile view [ref 1.2.1.2]
1.2.1.2.7	Change profile picture button
1.2.1.2.7.1	Upload new picture
1.2.1.2.7.2	Submit picture update
1.2.1.2.7.3	Return to profile view [ref 1.2.1.2]
1.2.1.3	Change password button
1.2.1.3.1	Taking current password
1.2.1.3.2	Taking new password
1.2.1.3.3	Confirming new password
1.2.1.3.4	Submitting the change
1.3	**Manage Staff Accounts**
1.3.1	View staff list in table format
1.3.2	Add staff button
1.3.2.1	Staff registration (name, email, username, password, phone, role)
1.3.2.2	Submit staff creation form

1.3.3	Edit staff button
1.3.3.1	Load existing staff info and allow changes
1.3.3.2	Submit updated staff info
1.3.4	Delete staff button
1.3.4.1	Confirm deletion prompt
1.3.4.2	Remove staff from database
1.4	**Inventory Management**
1.4.1	Add product button
1.4.1.1	Enter product name, category, price, quantity
1.4.1.2	Submit product form
1.4.2	Search product button
1.4.2.1	Enter product name or category
1.4.2.2	Submit search and display results in a table
1.4.3	Show all products button
1.4.3.1	Show product name, price, stock quantity
1.4.3.2	Edit product button
1.4.3.3	Delete product button with confirmation
1.5	**Sales Reporting & Data Export**
1.5.1	View daily sales report
1.5.3	View monthly sales report
1.5.4	Search transactions by date, user, or product

1.5.5	Export sales data to Excel/PDF
1.6	**Discounts & Configuration**
1.6.1	Set discount percentage for items or combos

1.6.2	Set tax rate
1.6.3	Set business hours
1.6.4	Set currency format
1.7	**Feedback Management**
1.7.1	View all customer feedback in table format
1.7.2	Mark feedback as reviewed
1.7.3	Reply to feedback if necessary
1.8	**Logout**
1.8.1	Return to login page

2.3.1.2 Staff Requirements

1.0	**User home page-All features availability**
1.1	**Login** [ref Admin 1.1]
1.2	**Dashboard**
1.2.1	Side menu
1.2.1.1	Dashboard button [ref 1.2]
1.2.1.2	View profile button [ref Admin 1.2.1.2]
1.2.1.3	Edit profile button [ref Admin 1.2.1.2.6]

1.2.1.4	Change profile picture button [ref Admin 1.2.1.2.7]
1.2.1.5	Change password button [ref Admin 1.2.1.3]
1.3	**Order Processing**
1.3.1	Add item to cart button
1.3.2	Edit cart item button
1.3.3	Remove cart item button
1.3.4	Apply discount (if authorized)
1.3.5	Calculate total and change
1.3.5	Calculate total and change
1.3.6	Generate invoice (print/digital)
1.3.7	Hold order button
1.3.8	Resume held order button
1.4	**Product Search**
1.4.1	Search product by name/category [ref Admin 1.4.2]
1.5	**Sales Monitoring**
1.5.1	View today's sales summary
1.5.2	View personal sales history
1.6	**Report Issues**
1.6.1	Flag low stock
1.6.2	Report wrong price or damaged product
1.7	**Logout**
1.7.1	Return to login page

2.3.1.3 Customer Requirements

1.0	**Place Order**
1.0.1	Select items from menu
1.0.2	Modify order before billing
1.0.3	Confirm order
1.1	**Receive Invoice**
1.1.1	Printed invoice option
1.1.2	Digital invoice option via SMS/email
1.2	**View Prices**
1.2.1	See price before payment
1.3	Request Refund/Cancellation
1.3.1	Request before order completion
1.3.2	Confirm with staff
1.4	**Feedback Submission**
1.4.1	Provide complaints or suggestions
1.4.2	Submit feedback to Admin

2.3.2 Non-Functional Requirements

1.0	**Usability**
1.0.1	The system should provide a clean, intuitive, and role-based interface (Admin, Staff, Customer).
1.0.2	All menus and options should be easy to navigate, with clear labels and consistent design.
1.1.3	Error messages should be meaningful and guide users toward corrective actions.
2.0	**Reliability**
2.0.1	The system should ensure data consistency during sales and inventory transactions.
2.0.2	The system must automatically back up sales and inventory data daily.
2.0.3	In case of system failure, recovery should be possible within 30 minutes using the backup.
3.0	**Security**
3.0.1	Secure authentication must be enforced for Admin and Staff logins.
3.0.2	Passwords must be stored in encrypted form.
3.0.3	Role-based access control must restrict Staff from accessing Admin- only features.
3.0.4	The system must lock a user account after 5 failed login attempts.
4.0	**Maintainability**
4.0.1	The system should follow a modular design to allow for easy updates.
4.0.2	Configuration files (tax rates, currency, business hours) must be editable without modifying source code.
5.0	**Portability**
5.0.1	The system must run on standard Windows-based desktops used in the AIUB canteen.

5.0.2	The system should be easily installable without requiring advanced technical knowledge.
6.0	**Performance**
6.0.1	Response Time
6.0.1	The system must process an order and generate an invoice within 3 seconds after confirmation.
6.0.2	Searching for products or transactions should return results within 2 seconds for up to 50,000 records.
7.0	**Availability**
7.0.1	The system should have up-time of at least 99% during business hours.
7.0.2	Scheduled maintenance downtime must occur outside canteen operating hours.
8.0	**Scalability**
8.0.1	The system must support at least 10 concurrent Staff users on POS terminals.
8.0.2	The system should handle up to 1,000 transactions per day without performance degradation.
9.0	**Data Storage**
9.0.1	The system must store at least 5 years of transaction history without performance issues.
9.0.2	Archived records must be retrievable within 10 seconds.

3. SOFTWARE DESIGN

3.1 System Design

Use Case Diagram

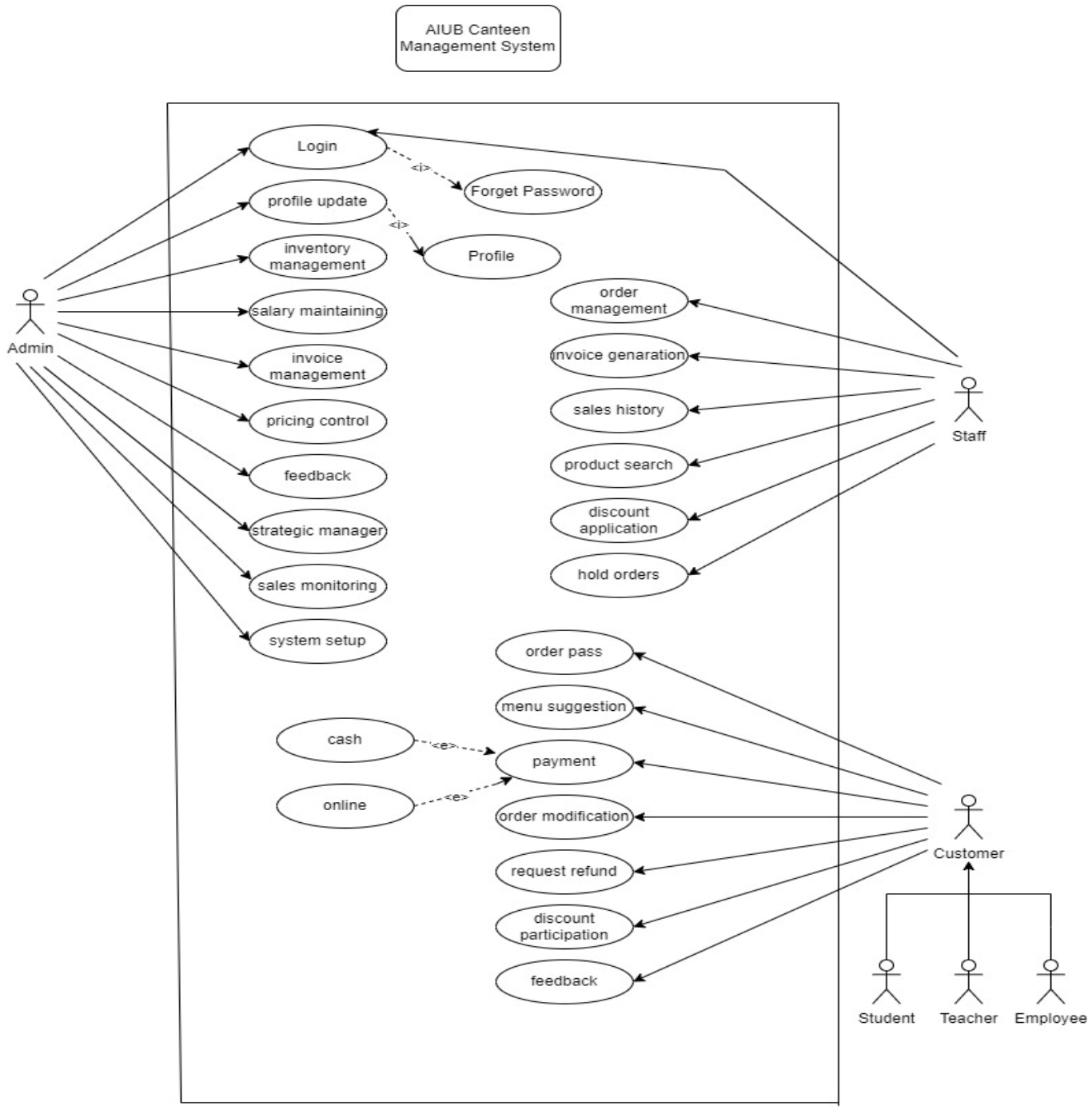

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Fig 2: Use Case Diagram

Class Diagram

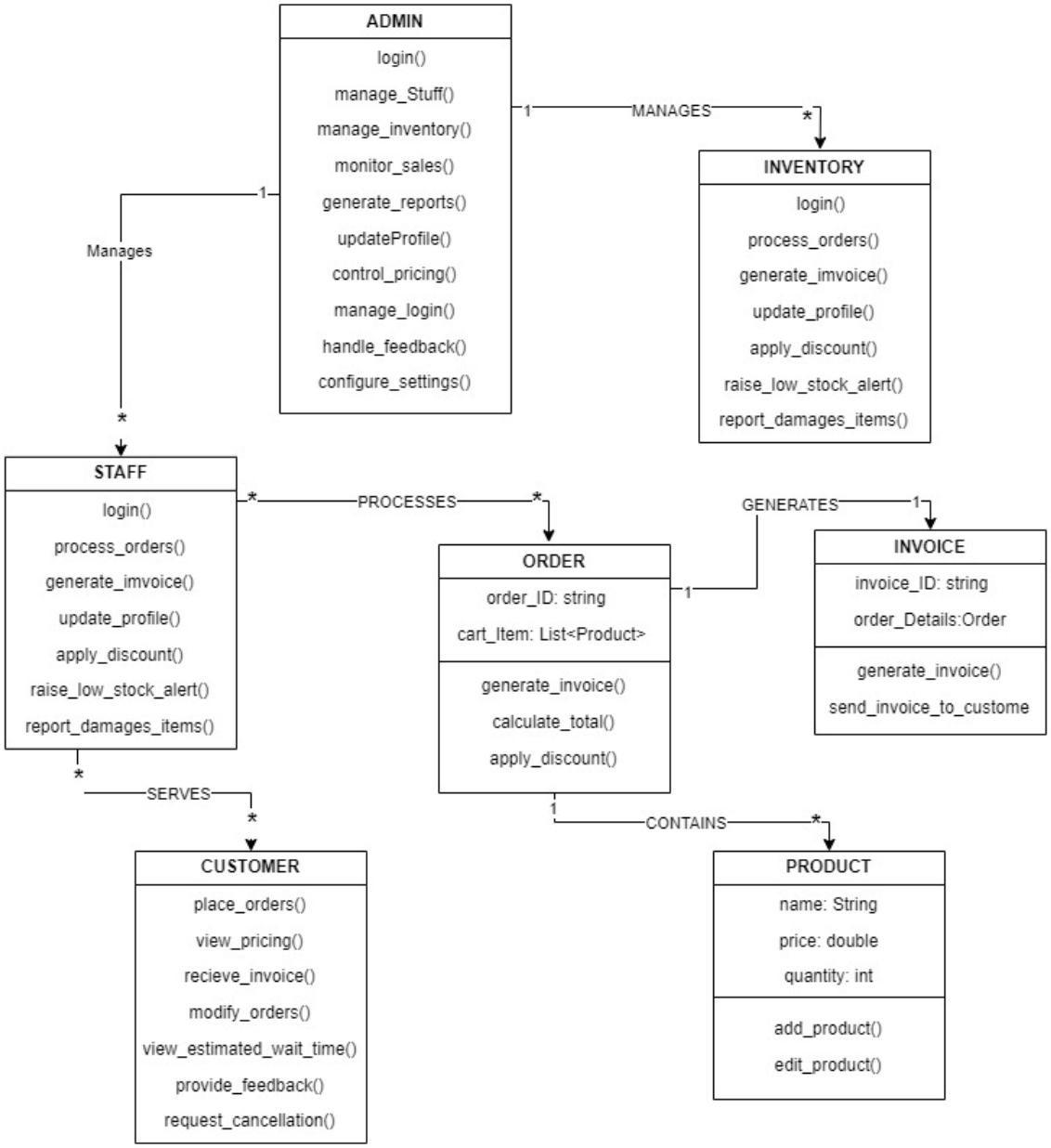

Fig 3: Class Diagram

Activity Diagram

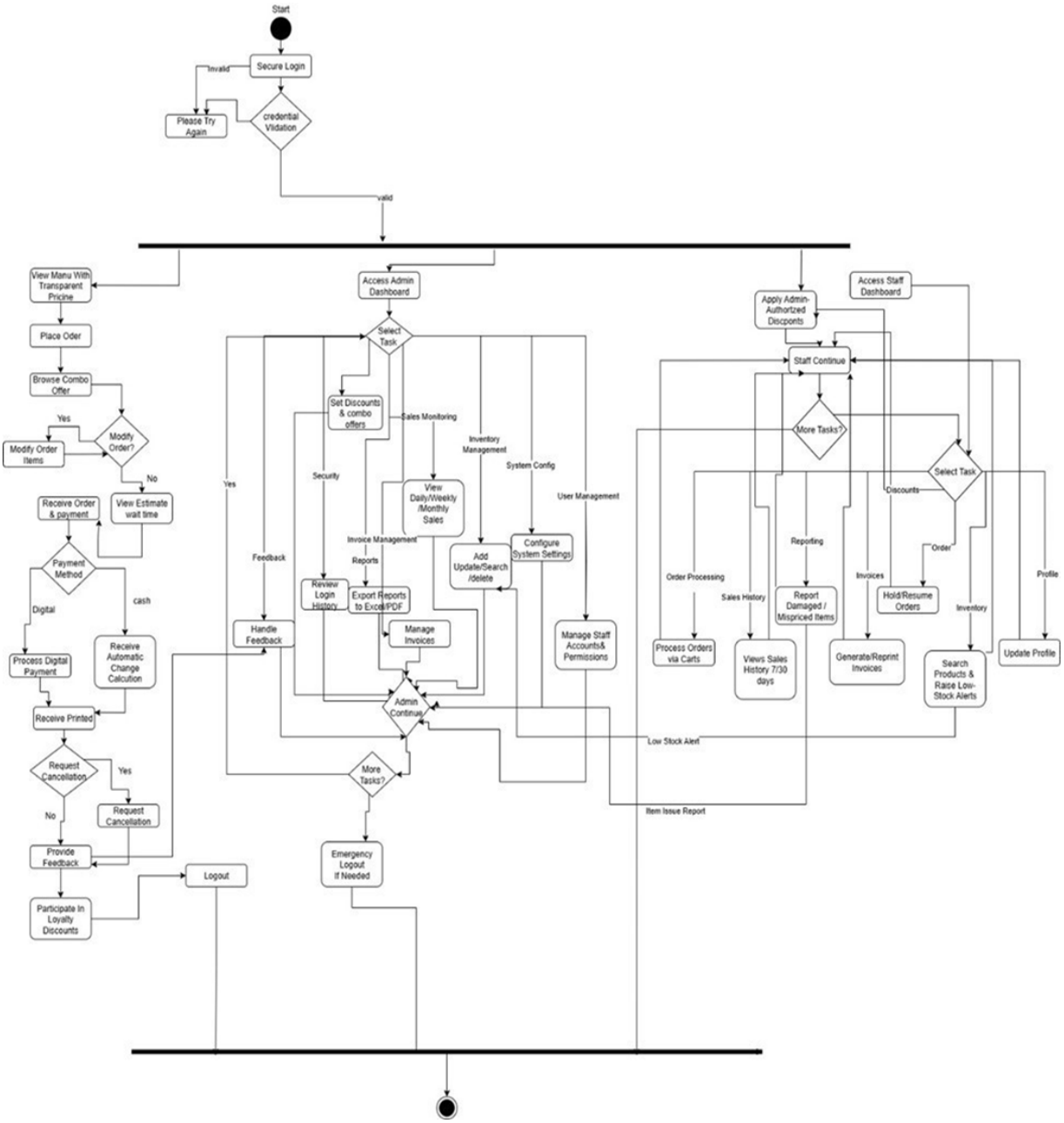

Fig 4: Activity Diagram

1.1 UI / Wireframe Design using Figma

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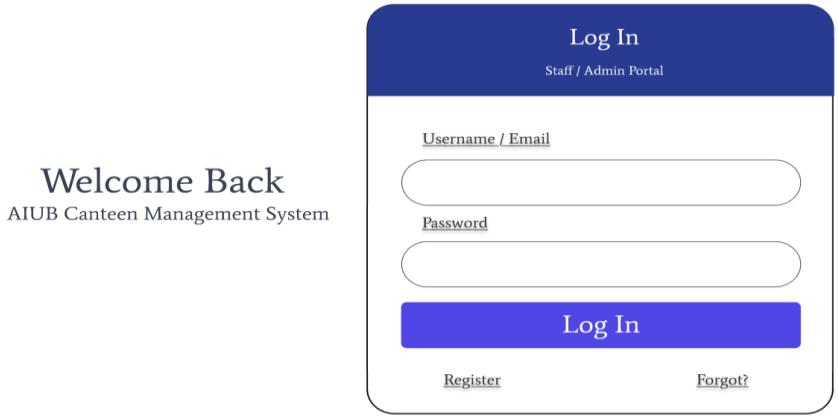
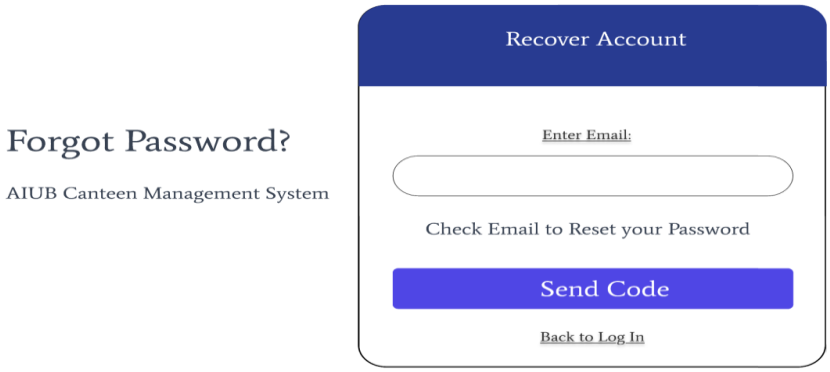
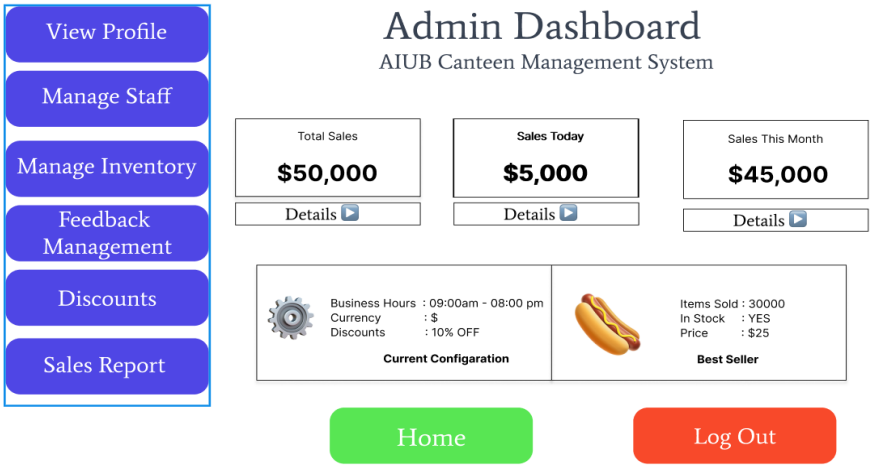
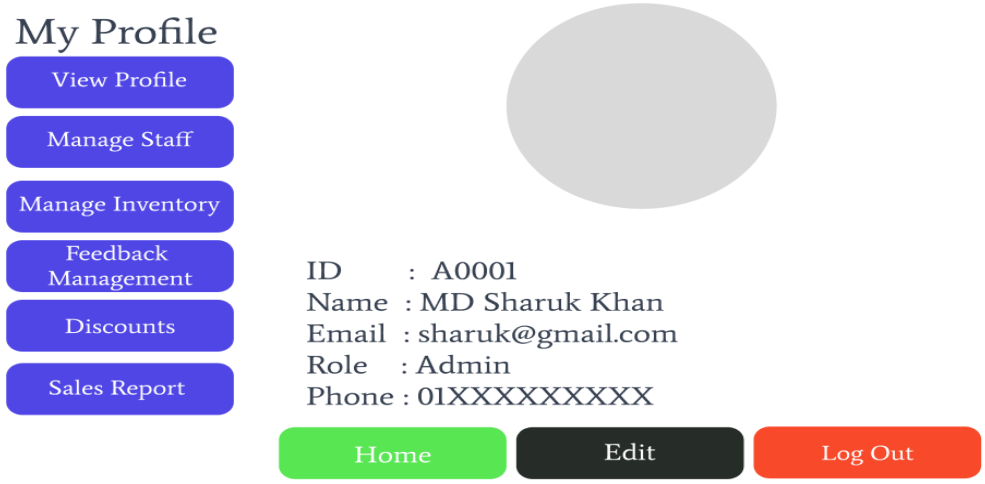
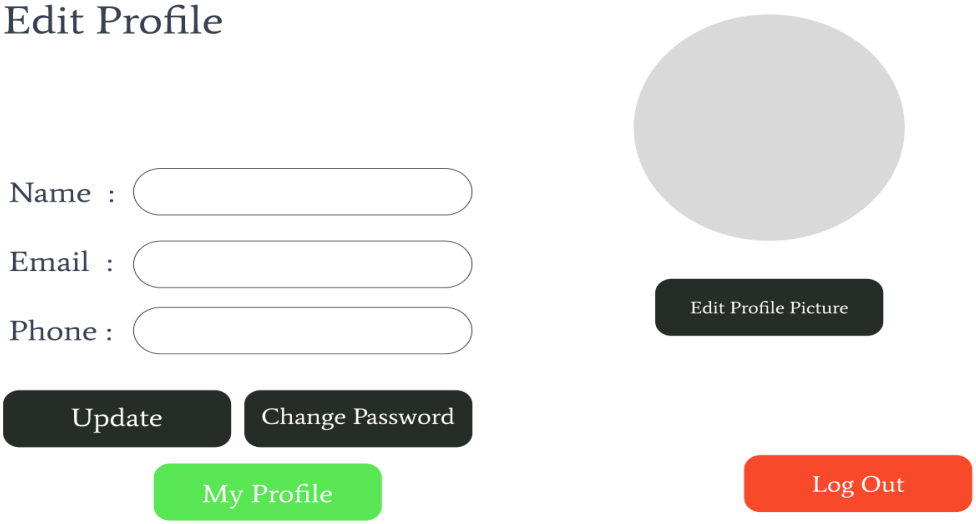
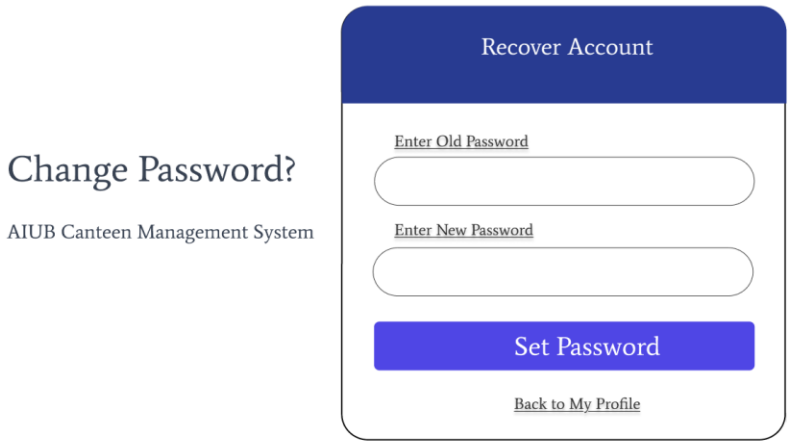







My Profile
View Profile
Manage Staff
Manage Inventory
Feedback Management
Discounts
Sales Report
ID : A0001
Name : MD Sharuk Khan
Email : sharuk@gmail.com
Role : Admin
Phone : 01XXXXXXXXX
Home
Edit
Log Out


Edit Profile
Name :
Email :
Phone :
Edit Profile Picture
Update
Change Password
My Profile
Log Out


Change Password?
AIUB Canteen Management System
Recover Account
Enter Old Password
Enter New Password
Set Password
Back to My Profile

Staff Registration

Name :

Email :

Username:

Password:

Phone:

Role:

Add Profile Picture

ADD

Cancel

Manage Staffs

Name	ID	Email	Phone	Role

Add Staff

View Staff

Delete

Return

Manage Product

View Profile

Manage Staff

Manage Inventory

Feedback Management

Discounts

Sales Report

Name :

Category:

Search

Name	ID	Quantity	Price

Add New

New Product

Name :

Price :

Quantity :

Category:

Save | Delete | Cancel

Registration

Name :

Email :

Username:

Password:

Phone:

Role:

Confirm

Add Profile Picture

Cancel

Status	Feedback Email	Feedback	
✅			Replay
✅			Replay
❌			Replay
✅			Replay
❌			Replay
✅			Replay

Log Out

Discounts & Configaration

View Profile
Manage Staff
Manage Inventory
Feedback Management
Discounts
Sales Report

Set Tax Rate:
Set Business Hours:
Set Currency Format:
Set Discount:

Save Settings

Home
Log Out

Date:
Transaction ID
User ID

Search

Name	ID	Quantity	Price

Daily Sales
Monthly Sales
Export to pdf
Return

View Profile
Order Processing
Product Search
Sales Monitoring
Report Issue

Welcome Back

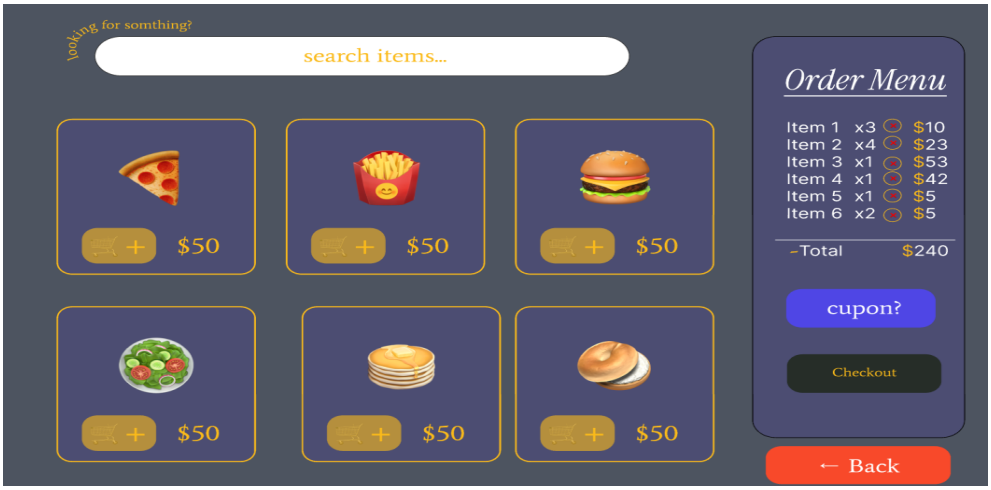
AIUB Canteen Management System

Log Out

Staff Profile

ID : S0001
Name : MD Ironman Khan
Email : ironman@gmail.com
Role : Staff
Phone : 01XXXXXXXXX

Home | Edit | Log Out



Search Product

Name :
Category:

Search

Name	ID	Quantity	Price

←Back

Report Issue

Start Typing Here...

Submit

Damaged Products

Low Stock

←Back

My Sales

Export to pdf

Total Sales	My Sales Today	My Sales This Month
$50,000	$5,000	$45,000

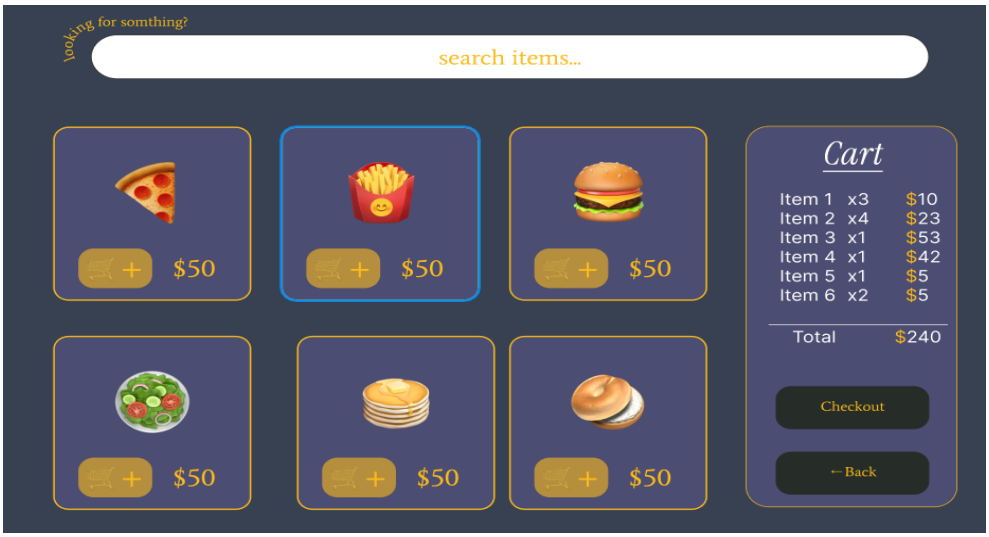
←Back

AIUB Canteen

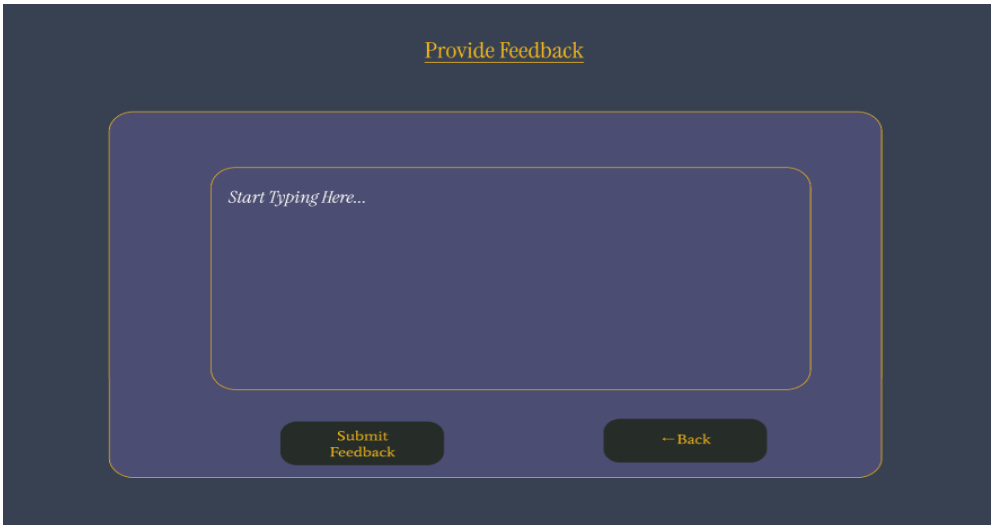
Welcome, Feeling Hungry ? — Browse Menu

Help us improve... — Provide Feedback

Billing Issues... — Request Refund


looking for somthing?
search items...
$50
$50
$50
$50
$50
$50
Cart
Item 1 x3 $10
Item 2 x4 $23
Item 3 x1 $53
Item 4 x1 $42
Item 5 x1 $5
Item 6 x2 $5
Total $240
Checkout
←Back

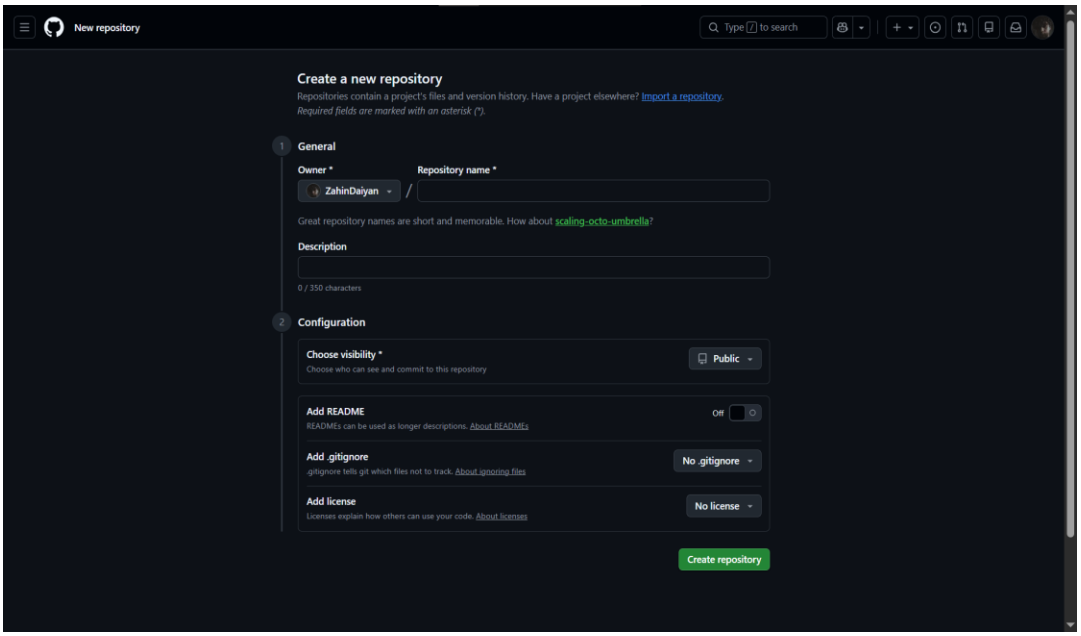
Confirm Transaction
Cart
Item 1 x3 $10
Item 2 x4 $23
Item 3 x1 $53
Item 4 x1 $42
Item 5 x1 $5
Item 6 x2 $5
Total $240
Print Invoice
Print Invoice
Confirm
←Back

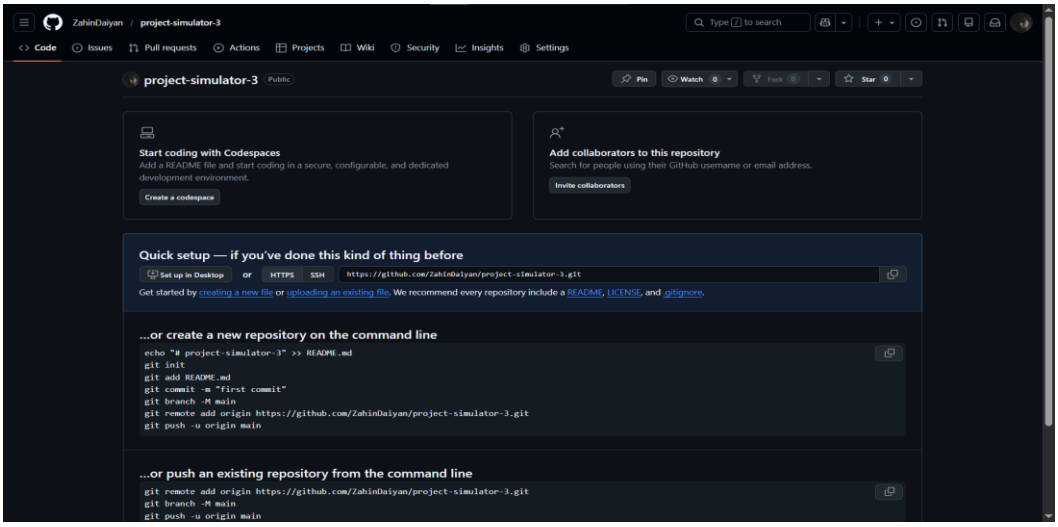

Provide Feedback
Start Typing Here...
Submit Feedback
←Back

4. GIT WORKFLOW

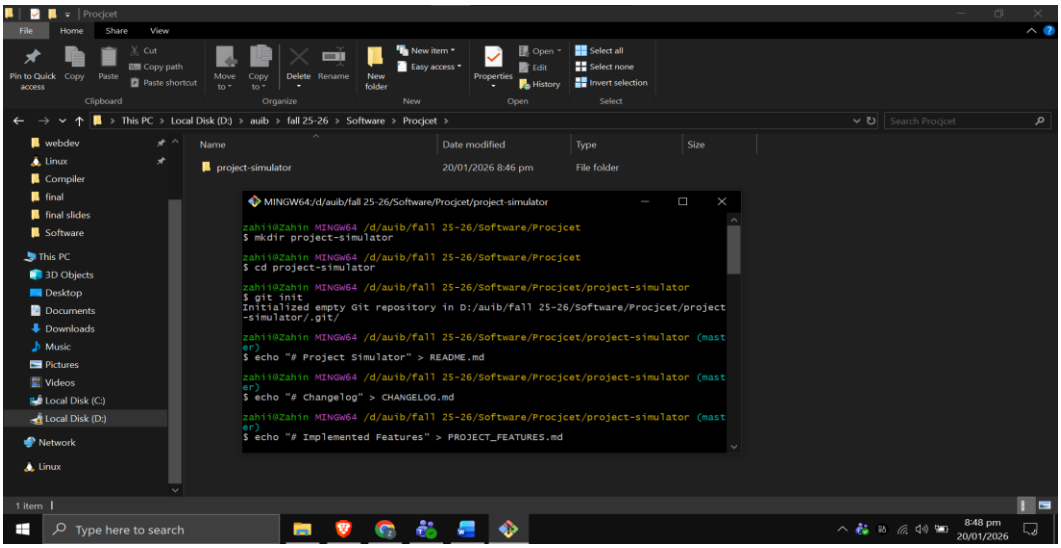
Github : https://github.com/ZahinDaiyan/project-simulator-3

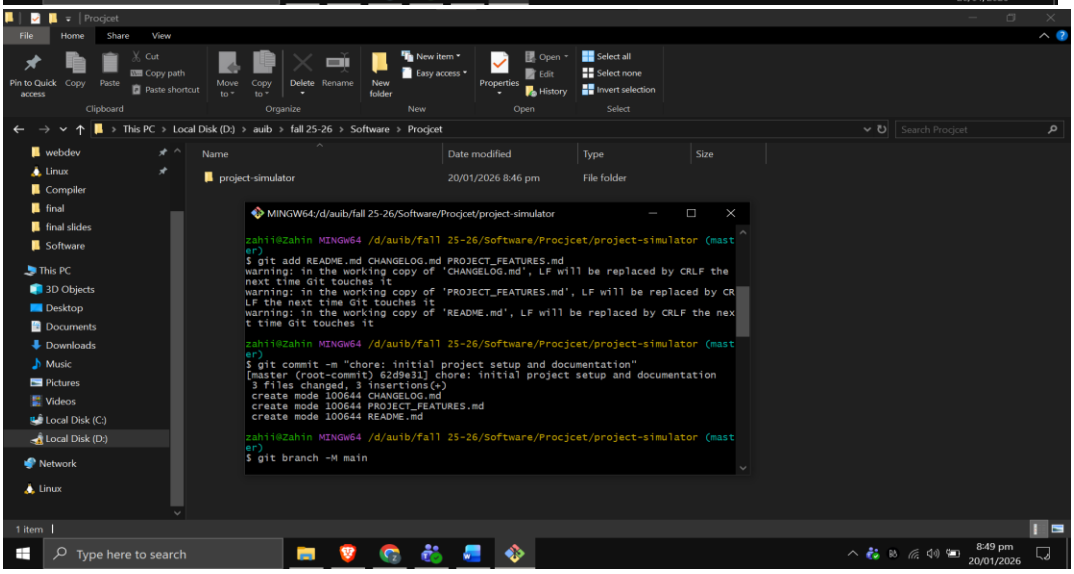
Created a new Repo

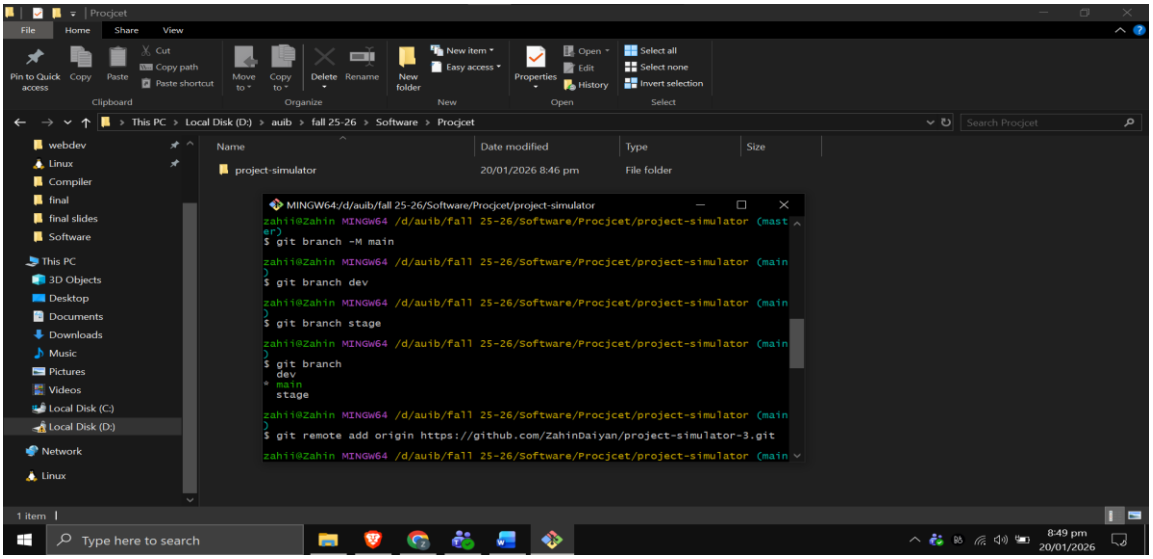


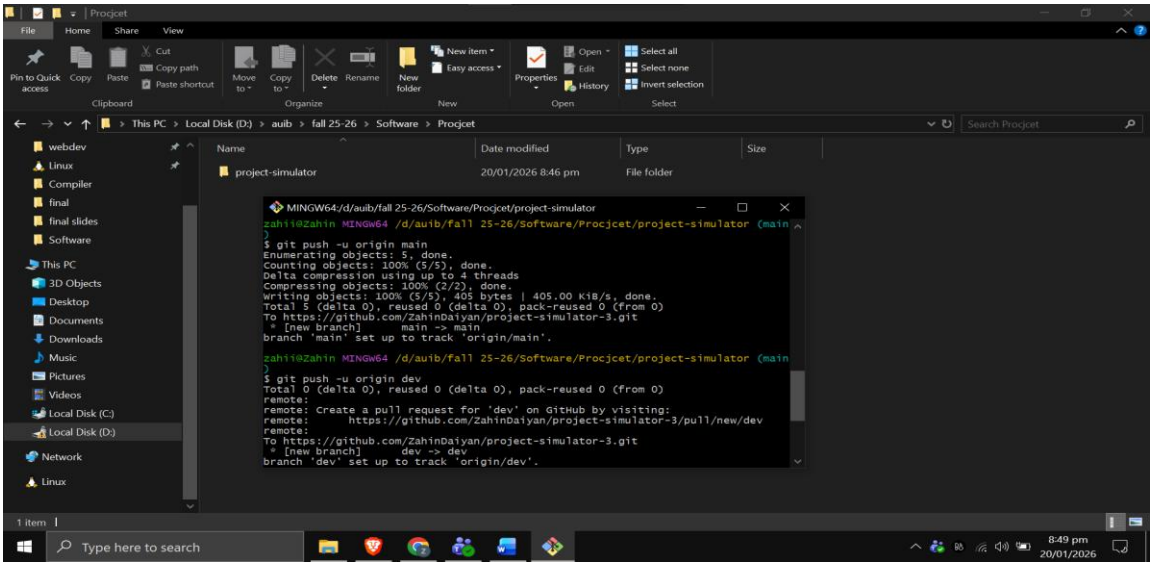


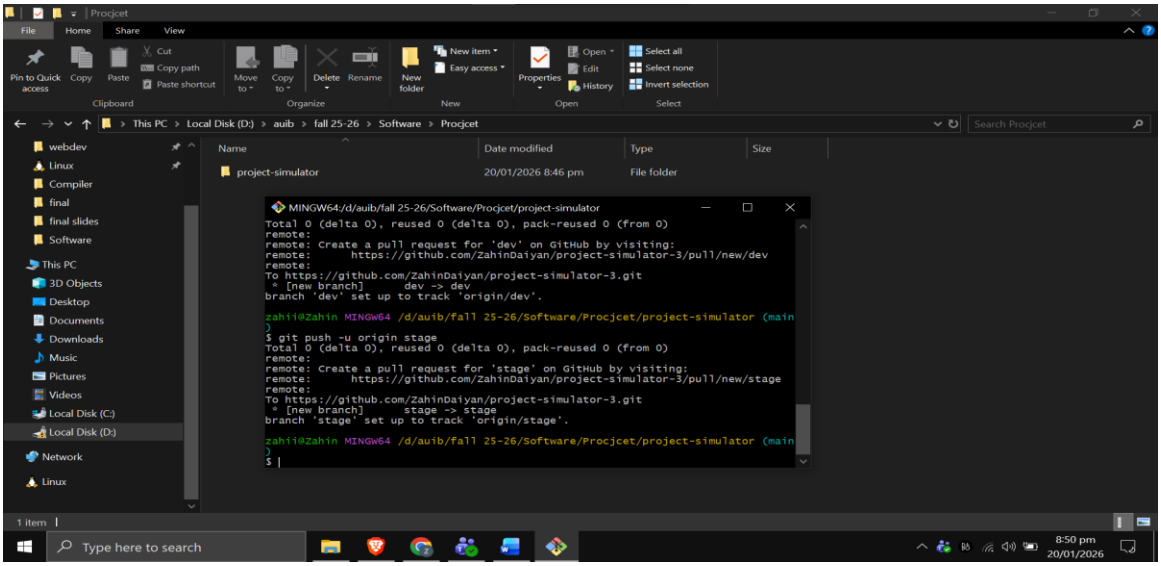
Create a Folder and Initial and create 3 files and branches

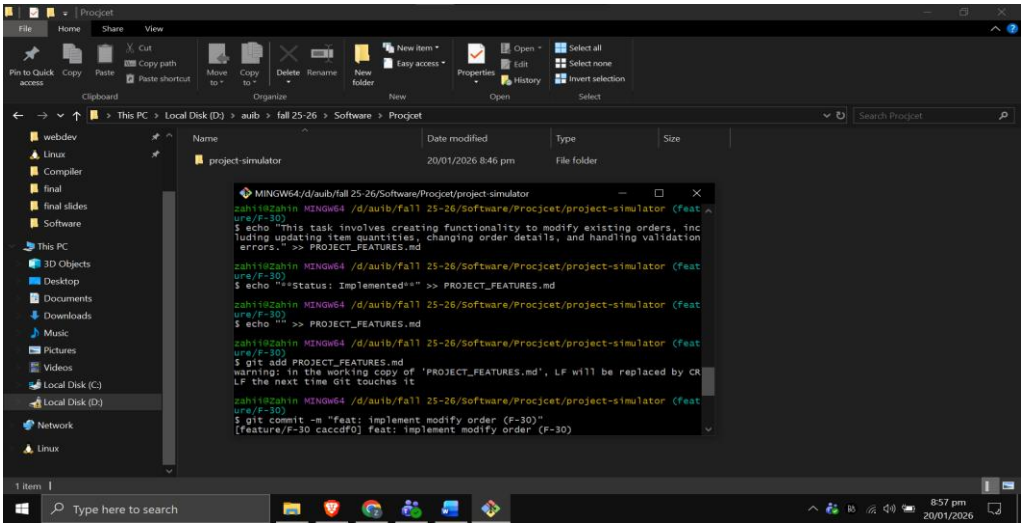


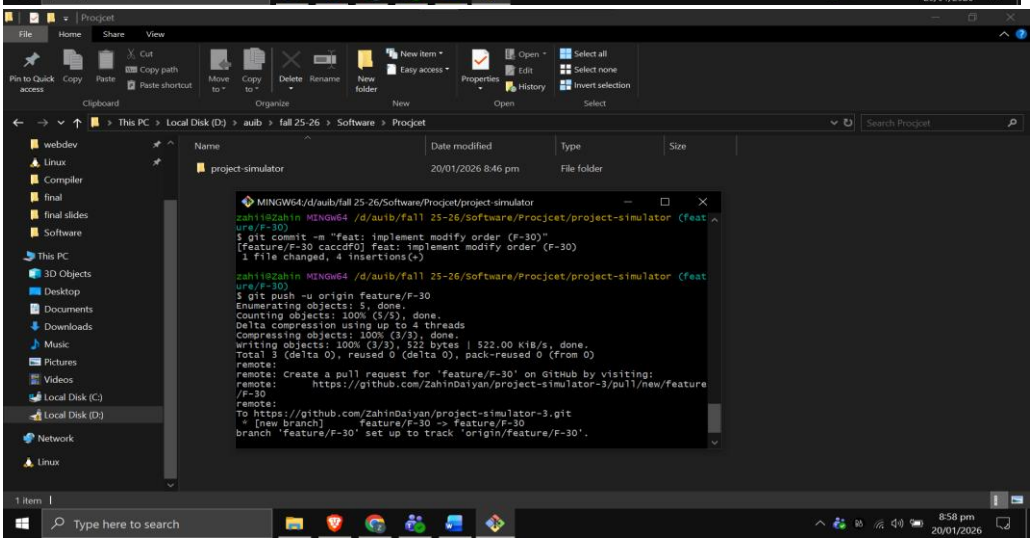



MINGW64:/d/auib/fall 25-26/Software/Procjcet/project-simulator
zahii@Zahin MINGW64 /d/auib/fall 25-26/Software/Procjcet/project-simulator (master)
$ git branch -M main
zahii@Zahin MINGW64 /d/auib/fall 25-26/Software/Procjcet/project-simulator (main)
$ git branch dev
zahii@Zahin MINGW64 /d/auib/fall 25-26/Software/Procjcet/project-simulator (main)
$ git branch stage
zahii@Zahin MINGW64 /d/auib/fall 25-26/Software/Procjcet/project-simulator (main)
$ git branch
dev
* main
stage
zahii@Zahin MINGW64 /d/auib/fall 25-26/Software/Procjcet/project-simulator (main)
$ git remote add origin https://github.com/ZahinDaiyan/project-simulator-3.git
zahii@Zahin MINGW64 /d/auib/fall 25-26/Software/Procjcet/project-simulator (main

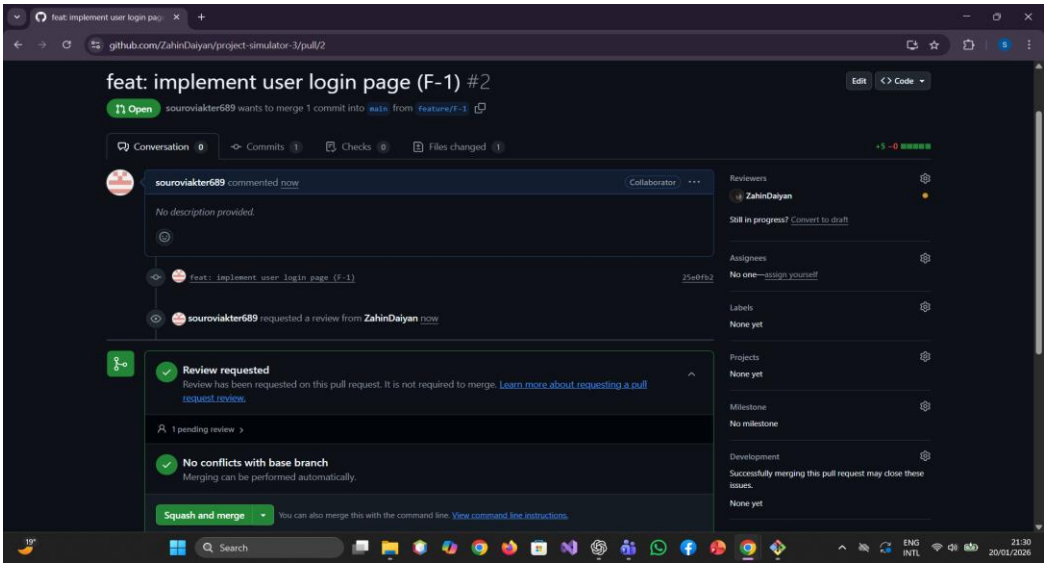

MINGW64:/d/auib/fall 25-26/Software/Procjcet/project-simulator
zahii@Zahin MINGW64 /d/auib/fall 25-26/Software/Procjcet/project-simulator (main)
$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (5/5), 405 bytes | 405.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ZahinDaiyan/project-simulator-3.git
* [new branch] main -> main
branch 'main' set up to track 'origin/main'.
zahii@Zahin MINGW64 /d/auib/fall 25-26/Software/Procjcet/project-simulator (main)
$ git push -u origin dev
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'dev' on GitHub by visiting:
remote: https://github.com/ZahinDaiyan/project-simulator-3/pull/new/dev
remote:
To https://github.com/ZahinDaiyan/project-simulator-3.git
* [new branch] dev -> dev
branch 'dev' set up to track 'origin/dev'.

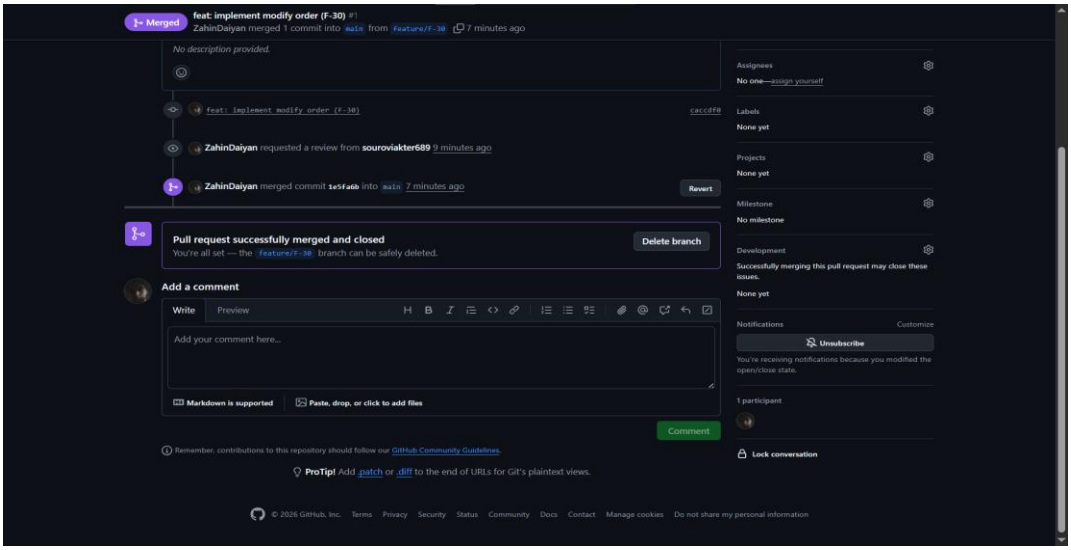

MINGW64:/d/auib/fall 25-26/Software/Procjcet/project-simulator
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'dev' on GitHub by visiting:
remote: https://github.com/ZahinDaiyan/project-simulator-3/pull/new/dev
remote:
To https://github.com/ZahinDaiyan/project-simulator-3.git
* [new branch] dev -> dev
branch 'dev' set up to track 'origin/dev'.
zahii@Zahin MINGW64 /d/auib/fall 25-26/Software/Procjcet/project-simulator (main)
$ git push -u origin stage
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'stage' on GitHub by visiting:
remote: https://github.com/ZahinDaiyan/project-simulator-3/pull/new/stage
remote:
To https://github.com/ZahinDaiyan/project-simulator-3.git
* [new branch] stage -> stage
branch 'stage' set up to track 'origin/stage'.
zahii@Zahin MINGW64 /d/auib/fall 25-26/Software/Procjcet/project-simulator (main)
$

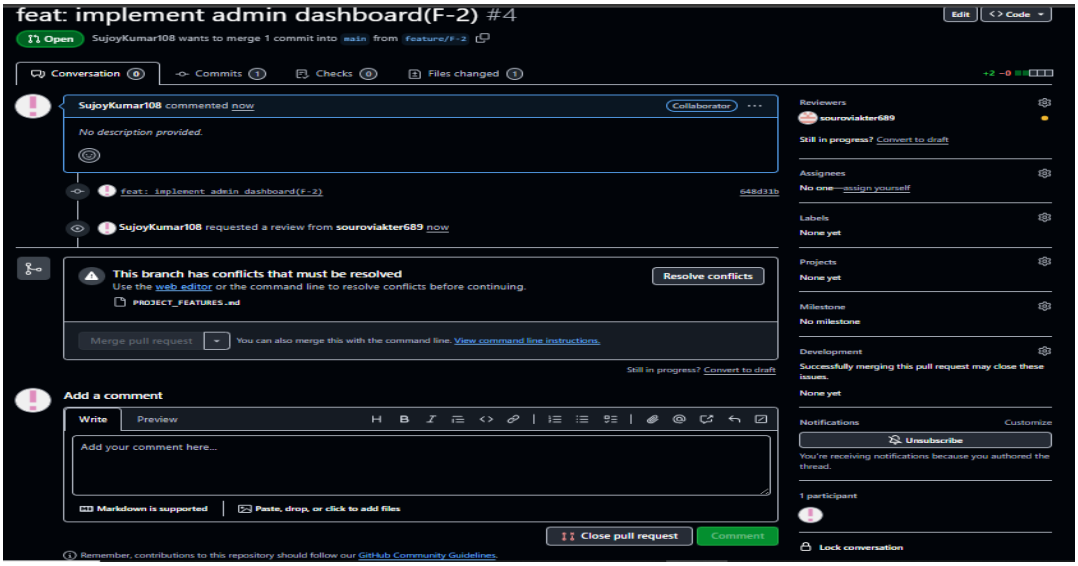


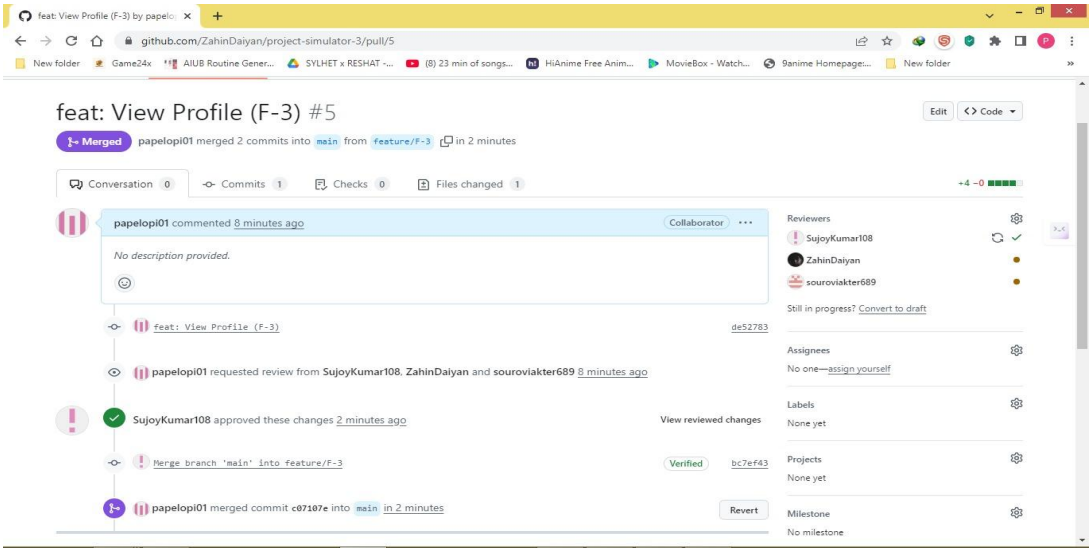


Make a Peer Review Request



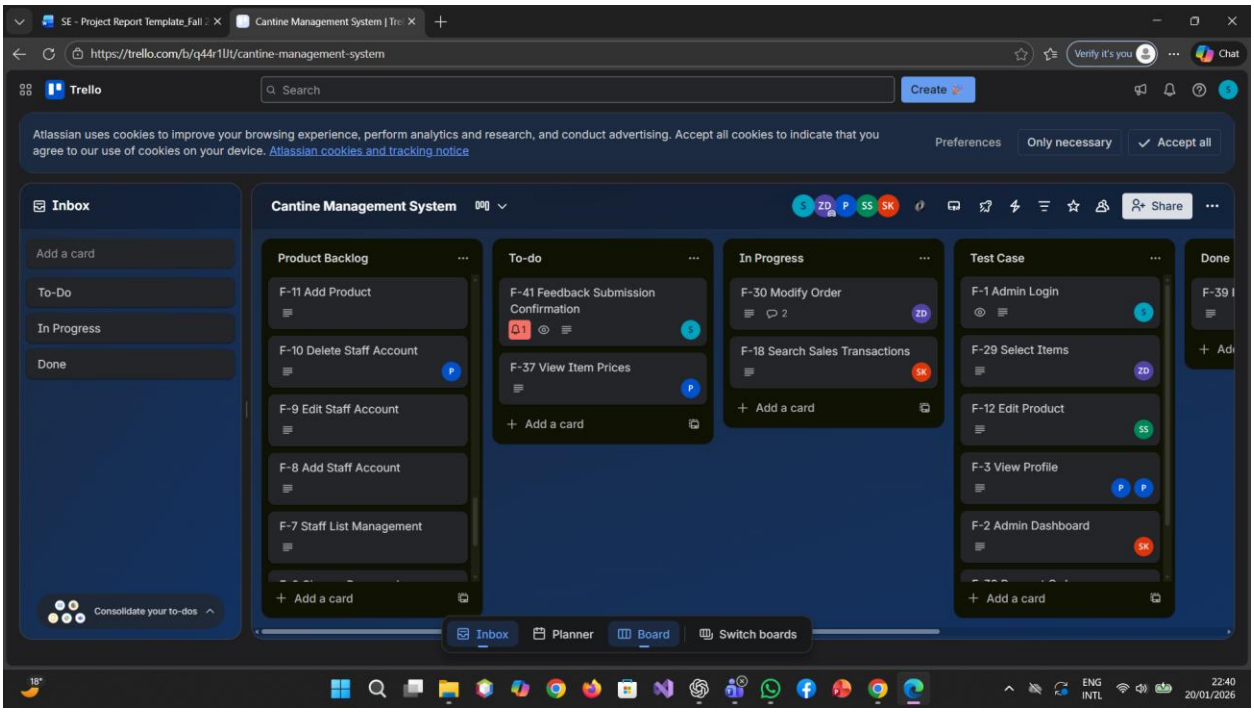

Merged
feat: implement modify order (F-30) #1
ZahinDaiyan merged 1 commit into main from feature/F-30 7 minutes ago
No description provided.
feat: implement modify order (F-30)
caccdf0
ZahinDaiyan requested a review from souroviakter689 9 minutes ago
ZahinDaiyan merged commit 1e5fa6b into main 7 minutes ago
Revert
Pull request successfully merged and closed
You're all set — the feature/F-30 branch can be safely deleted.
Delete branch
Add a comment
Write
Preview
Add your comment here...
Markdown is supported
Paste, drop, or click to add files
Comment
Remember, contributions to this repository should follow our GitHub Community Guidelines.
ProTip! Add .patch or .diff to the end of URLs for Git's plaintext views.
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
None yet
Notifications
Customize
Unsubscribe
You're receiving notifications because you modified the open/close state.
1 participant
Lock conversation
© 2026 GitHub, Inc. Terms Privacy Security Status Community Docs Contact Manage cookies Do not share my personal information


feat: implement admin dashboard(F-2) #4
Edit
Code
Open
SujoyKumar108 wants to merge 1 commit into main from feature/F-2
Conversation 0
Commits 1
Checks 0
Files changed 1
+2 −0
SujoyKumar108 commented now
Collaborator
No description provided.
feat: implement admin dashboard(F-2)
648d31b
SujoyKumar108 requested a review from souroviakter689 now
This branch has conflicts that must be resolved
Use the web editor or the command line to resolve conflicts before continuing.
PROJECT_FEATURES.md
Resolve conflicts
Merge pull request
You can also merge this with the command line. View command line instructions.
Still in progress? Convert to draft
Add a comment
Write
Preview
Add your comment here...
Markdown is supported
Paste, drop, or click to add files
Close pull request
Comment
Remember, contributions to this repository should follow our GitHub Community Guidelines.
Reviewers
souroviakter689
Still in progress? Convert to draft
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
None yet
Notifications
Customize
Unsubscribe
You're receiving notifications because you authored the thread.
1 participant
Lock conversation


feat: View Profile (F-3) by papelo
github.com/ZahinDaiyan/project-simulator-3/pull/5
New folder
Game24x
AIUB Routine Gener...
SYLHET x RESHAT -...
(8) 23 min of songs...
HiAnime Free Anim...
MovieBox - Watch...
9anime Homepage:...
New folder
feat: View Profile (F-3) #5
Edit
Code
Merged
papelopi01 merged 2 commits into main from feature/F-3 in 2 minutes
Conversation 0
Commits 1
Checks 0
Files changed 1
+4 −0
papelopi01 commented 8 minutes ago
Collaborator
No description provided.
feat: View Profile (F-3)
de52783
papelopi01 requested review from SujoyKumar108, ZahinDaiyan and souroviakter689 8 minutes ago
SujoyKumar108 approved these changes 2 minutes ago
View reviewed changes
Merge branch 'main' into feature/F-3
Verified
bc7ef43
papelopi01 merged commit c07107e into main in 2 minutes
Revert
Reviewers
SujoyKumar108
ZahinDaiyan
souroviakter689
Still in progress? Convert to draft
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone

Trello Board

https://trello.com/invite/b/6919346690fb593c7f4944cc/ATTIde64e0b3049ddef0dd20b3be30170e3146376FE0/cantine-management-system



5. SOFTWARE TESTING

Goal of Testing

The purpose of testing in the AIUB Canteen Management System project is to verify that the proposed solution provides robust high-quality software and meets all the requirements stipulated accompanying this document and assures reliable functioning of canteens in AIUB. As this application aims to substitute prior manual processes to an automated environment, the aspect of testing becomes even more important in order fix all obvious bugs on a secure login process, inventory control cash register cashier billings invoice issuing reporting feed-back system. Testing ensures that every feature works fine, and the links between different modules are smooth.

Apart from catching mistakes, testing verifies that the system delivers value to its end-users. For your administrators, you need transparency and control; for staff efficient task-oriented utility; and for customers an unblemished delivery of service. Testing validates such varied user requirements and separately proves non-functional properties like security, performance, and usability. Without rigorous testing, the system may not instill confidence with stakeholders and may be adopted poorly or there is a risk of running against operations. In a nutshell, the purpose of testing is twofold; to check conformance to requirements (is the system doing what it is supposed? and for quality and reliability (how well does the system do it in practice?). This will be performed by employing successive testing strategies in various stages and steps of the project.

Testing Methods and Their Application

Unit Testing

Unit testing focuses on testing the smallest parts of the system, typically at the level of functions, classes, or individual modules. The aim is to verify that each unit behaves correctly in isolation before it is combined with others. This reduces the complexity of debugging, since if a defect is found, it is easier to pinpoint its source in a single unit rather than across the whole system.

Unit testing is often automated to ensure repeated and consistent verification whenever the code is updated. In the AIUB Canteen Management System, unit testing will be performed for main functions such as login validation, stock update, bill calculate, and sales history retrieve. To do that, we would want to test data like valid vs invalid username & password combinations as well as empty fields and the like. Likewise, in billing, the system will be tested for varying discounts on various combinations of taxes on a variety of quantity, etc. and mention that the totals are correct. These tests provide certainty that the core logic of the system is right. Unit testing is important here, as we deal with financial transactions and related functions for security. A minor mistake in the sum of invoices, or a problem while logging in, would destroy trust. Unit testing does that by making sure a small piece works well alone, so you can trust it while building a complex system.

Integration Testing

Integration testing validates how different modules or subsystems interact with one another once they are combined. Even if each unit works correctly on its own, errors can still occur when they exchange data. This type of testing ensures that the system functions as a cohesive whole by checking interfaces, data flows, and dependencies. Integration testing can be incremental (testing pairs of modules gradually) or big bang (testing the entire system integration at once).

In our canteen system, the Staff POS module will be tested in an integrated environment with the Inventory to verify logically that they interact as expected. If an employee makes a sale, then the number of products should be decreased from quantity in the inventory database. Another end-to-end test will ensure that an Admin created discount is correctly applied at the Staff POS for billing.

Likewise, the mapping of reporting module and database will also verify whether reports generated by Admin correctly represent orders staff have placed.

The integration testing is very important in the project as it maintains data integrity among various modules. If a sale doesn't offset inventory, or if discounts are not properly considered, the system may generate inaccurate reports or worse yet, financial errors. A smooth integration is necessary to ensure information consistency and integrity between user roles (Admin, Staff and Customer).

System Testing

System testing evaluates the entire integrated system. Unlike unit and integration testing, which focus on specific components or interactions, system testing validates the complete software

against the original requirements. It includes both functional tests (does the system behave as specified?) and non-functional tests (does it perform well, is it secure, is it user-friendly?).

Regarding the system testing of the AIUB Canteen Management System, it will perform in real life. Customer orders, the employees process it, there is an invoice created and sometime after Admin reviews a sales report. This is a simulation of the complete order-to-record cycle. Some of the non-functional aspects that will be tested are performance during the busiest hours in some cafeteria during lunch times to verify if concurrent transactions can be performed on a timely fashion, and security to ensure that employees cannot use admin-related features.

System testing is critical, as it ensures the complete system meets its objectives. And because the system must do many things for different purposes, testing guarantees all processes - from ordering to reporting- work seamlessly and well.

Smoke Testing

Smoke testing is a rapid and surface-level kind of testing carried out on both new builds, and integration. Its aim is to validate whether fundamental and critical capabilities of the system work, and the software is stable enough for more testing. Consider it as a “sanity check”—if the system fails a smoke test, then any deeper tests have no value performing until issues are addressed.

Smoke testing would be the first thing we reach whenever this is developed for a project. E.g. Youve hooked together login and POS modules Your first smoke test should see if Admin can log in, or Staff can log in and try to buy a chocolate bar without your whole system dying. If these crucial elements operate satisfactorily, the building is considered safe, and more extensive tests are performed. Smoke testing is important in our project because it saves time and effort. By quickly detecting whether a build is stable, it prevents wasted resources from testing broken software. This is especially relevant in an academic project where resources are limited, and development time is tight.

Alpha Testing

Alpha testing is the initial phase of software testing where testers work with developers and QA team internally to ensure that the product is ready for release to customers. Its goal is to simulate real-world use in a controlled environment to test bugs, missing features, and usability issues. Alpha testing is normally conducted in several rounds, with the product being updated following each one until it achieves a stable and feature complete state.

In the canteen system, alpha testing will be carried out by the team where staff are admin, customers, and staff. for instance, if a tester is testing as Staff, he'll 'process order' and then work out of generated invoice view that the inventory update auto-populates.

This will then generate a report for another tester who is administering that sale to confirm that you are assigned to the correct linked sale. These controlled simulations aid in the

systemic detection of functionality and usability flaws before the software is distributed to real-world (canteen) users.

Alpha testing is important for our project because it acts as the last internal checkpoint before releasing the system to real users. By catching errors early in a controlled environment, it reduces the likelihood of users encountering major issues during beta testing or deployment.

Beta Testing

Beta testing is when we give out close-to-finished software's to real-end users and let them test it in a real setting. Vice alpha testing, which is done under the supervision of developers, beta testing offers customer feedback from actual serious users in the field. It identifies usability issues, edge cases, and real-world performance problems that may have been overlooked.

Beta testing in this project will run to specific canteen staff and very few customers inside AIUB. Staff will make real orders utilizing the POS, while Admins will run reports and see how it is doing. Clients will share their feedback on invoice transparency, transaction speed, and that of the service overall. This stage will expose usability problems such as difficulty in managing peak-hour wait lists or confusion in UI design. Beta testing is essential because it confirms that the system works in its natural habitat. The project prevents purely technically correct, but impractical, user-unfriendly and non-accepted products by involving the users early during development before releasing a final product.
AIUB Canteen Management System will be tested thoroughly at various levels using unit, integration, system, smoke alpha and beta testing. With unit and integration testing, you will be assured that the fundamental components of the system, and how they interact will work as expected. System & Smoke Testing The goal here is to ensure the entire system flows and is stable. Lastly, alpha and beta testing will indicate if the system is simple to use and performs for real users.

Test Case

Project Name: AIUB Canteen Management System			**Test Designed by: Sourovi Akter Purnima**	
Test Case ID: TC_1			**Test Designed date: 15-01-2026**	
Test Priority (Low, Medium, High): Medium			**Test Executed by: Sourovi Akter Purnima**	
Module Name: Login Session			**Test Execution date: 19-01-2026**	
Test Title: Verify login with valid username and password				
Description: Test the website login page				
Precondition: The user has a valid username and password				
Dependencies: User has a registered account.				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Go to the site 2. Enter username 3. Enter password 4. Click submit	**Username : Sourovi88 Password: 568**	**The user should log in to the application**	**As expected,**	**Pass**

Project Name: AIUB Canteen Management System	**Test Designed by: Sourovi Akter Purnima**
Test Case ID: TC_2	**Test Designed date: 15-01-2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: Sourovi Akter Purnima**
Module Name: Dashboard Entry Session	**Test Execution date: 19-01-2026**
Test Title: Verify entering Dashboard shows all required components	
Description: After login (or clicking Dashboard), the Dashboard page should load and display all listed components/controls (profile info panel, action buttons, and staff management area).	
Precondition: User is logged in with sufficient role (Admin) to see staff management.	
Dependencies: Log in with a valid Admin account.	

Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Log in as Admin & open Dashboard 2. Check all listed features 3. Verify all the listed components/controls	**Username: adminA Password: xyz**	**After logging in, the user loads the Dashboard without errors, and all components are visible, properly working and labeled, and clickable.**	**As expected,**	**Pass**

Project Name: AIUB Canteen Management System	**Test Designed by: Sourovi Akter Purnima**
Test Case ID: TC_3	**Test Designed date: 15-01-2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: Sourovi Akter Purnima**
Module Name: Edit Profile Session	**Test Execution date: 19-01-2026**
Test Title: Verify users can edit profile information	
Description: Ensure users can update their profile details.	
Precondition: User is logged in with valid credentials.	
Dependencies: Profile must exist.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to Dashboard 2. Click On "Edit Profile" button 3. Update name and phone number 4. Click submit	New Name: Purnima New Phone: 01289	Profile should be updated and returned to profile view with updated info shown.	As expected,	Pass

Project Name: AIUB Canteen Management System	**Test Designed by: MD Zahin Daiyan**
Test Case ID: TC_4	**Test Designed date: 16-01-2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: MD Zahin Daiyan**
Module Name: Staff Management Session	**Test Execution date: 19-01-2026**
Test Title: Verify new staff can be added successfully	
Description: Ensure admin can add a staff account through the dashboard.	
Precondition: Admin user is logged in.	
Dependencies: Staff management feature enabled.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to Dashboard 2. Click on "Manage Staff Accounts" 3. Click "Add Staff" 4. Fill in staff registration form (name, email, username, password, phone, role) 5. Submit form	Name: Zahin Email: Zahin@aiub.edu Username: Zahin123 Password: abc@123 Phone: 01289 Role: Staff	New staff should be added to the database and visible on the staff list table.	As expected,	Pass

Project Name: AIUB Canteen Management System	**Test Designed by: MD Zahin Daiyan**
Test Case ID:TC_05	**Test Designed date: 16-01-2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: MD Zahin Daiyan**
Module Name: Reporting & Data Management	**Test Execution date: 19-01-2026**
Test Title: Verify admin can export sales reports to Excel	
Description: Test the export functionality for admin sales report.	
Precondition: Admin logged in; sales data available.	
Dependencies: Sales report module active.	

Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
5. Go to "Reports" page **6. Select "Weekly Sales Report"** **7. Click "Export to Excel"**	**Range: last 7 days**	**Report generated loads successfully**	**As expected,**	**Pass**

Project Name: AIUB Canteen Management System	**Test Designed by: MD Zahin Daiyan**
Test Case ID:TC_6	**Test Designed date: 16-01-2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: MD Zahin Daiyan**
Module Name: Inventory Management	**Test Execution date: 19-01-2026**
Test Title: Verify that admin can add a new product	
Description: Admin should be able to add new products into inventory.	
Precondition: Admin logged into the system.	
Dependencies: Database connection active.	

Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
4. Go to Inventory Module **5. Click "Add Product** **6. Enter details** **7. Click Save**	**Product: Pizza, Price: 200, Stock: 50** **Product successfully added to inventory**	**Inventory page opens** **Add Product form opens** **Accepted Product successfully added to inventory**	**Product not added to inventory Error message displayed: "Database error"**	**Fail**

Project Name: AIUB Canteen Management System	**Test Designed by: Sujoy Kumar Mondol**
Test Case ID:TC_7	**Test Designed date: 17/01/2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: Sujoy Kumar Mondol**
Module Name: Customer Interaction	**Test Execution date: 20/01/2026**
Test Title: Customer requests invoice copy via Email	
Description: Ensure system sends invoices to customer's email.	
Precondition: Customer orders are completed. Email address is available.	
Dependencies: Email service configured.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
5. Staff completes sale 6. Customer requests invoice via email 7. System sends invoice	Order: Sandwich + Coke = 120 BDT Email: Sujoy@123	Order processed Request accepted Invoice copy received in customer's email	Invoice not received in customer's email	Fail

Project Name: AIUB Canteen Management System	**Test Designed by: Sujoy Kumar Mondol**
Test Case ID:TC_8	**Test Designed date: 17/01/2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: Sujoy Kumar Mondol**
Module Name: Customer Interaction	**Test Execution date: 20/01/2026**
Test Title: Verify that customer can cancel an order before final billing	
Description: Customers should be able to cancel an order before invoice generation.	
Precondition: Staff logged in and cart items selected.	
Dependencies: Sales Processing Module.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
6. Staff adds items to cart 7. Customer requests cancellation 8. Click "Cancel Order"	Product: Sandwich, Coke	Cart updated Cancellation request accepted Cart cleared, no invoice generated	Cancellation failed — "Cancel Order" button returned error. Cart not cleared, invoice generated. Defect logged.	Fail

Project Name: AIUB Canteen Management System	**Test Designed by: Sujoy Kumar Mondol**
Test Case ID: TC_9	**Test Designed date: 17/01/2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: Sujoy Kumar Mondol**
Module Name: Cart Management	**Test Execution date: 20/01/2026**
Test Title: Verify product can be added to cart	
Description: Test that a Staff can add a product to the cart	
Precondition: The Staff has a valid username and password	
Dependencies: Product must exist in inventory	

Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Go to product list **2. Select product** **3. Click "Add to Cart"**	**Product Name: Chicken Burger** **Quantity: 2**	**Product is added to cart with correct quantity and total**	**As expected,**	**Pass**

Project Name: AIUB Canteen Management System	**Test Designed by: A.K.M Deenat Papel**
Test Case ID: TC_10	**Test Designed date: 11/01/2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: A.K.M Deenat Papel**
Module Name: Invoice Generating	**Test Execution date: 20/01/2026**
Test Title: Verify invoice is generated after order confirmation	
Description: Test system generates invoice for completed order	
Precondition: User has items in cart and confirms order	
Dependencies: Printer (for hard copy) or email/SMS system (for soft copy)	

Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Confirm order **2. Select invoice** **Option (print/email) 3. Complete payment**	**Product: Chicken Burger (2) Total: 300 BDT**	**System generates invoice with order details, date, total, payment method**	**As Expected,**	**Pass**

Project Name: AIUB Canteen Management System	**Test Designed by: A.K.M Deenat Papel**
Test Case ID: TC_11	**Test Designed date: 11/01/2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: A.K.M Deenat Papel**
Module Name: Inventory Management	**Test Execution date: 20/01/2026**
Test Title: Verify Admin can update product quantity	
Description: Test system allows Admin to update inventory stock	
Precondition: Admin is logged in	
Dependencies: Product already exists in inventory	

Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Login as Admin **2. Go to Dashboard** **3. Click on "Manage Product"** **4. Search for product** **5. Update quantity** **6. Save changes**	**Product: Chicken Burger** **Old Qty: 50** **New Qty: 100**	**Product quantity updates successfully and reflects in inventory list**	**As Expected,**	**Pass**

Project Name: AIUB Canteen Management System	**Test Designed by: A.K.M Deenat Papel**
Test Case ID: TC_12	**Test Designed date: 11/01/2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: A.K.M Deenat Papel**
Module Name: Logout Session	**Test Execution date: 20/01/2026**
Test Title: Verify user can log out successfully	
Description: Test that user session ends after logout	
Precondition: User is logged in (Admin or Staff)	
Dependencies: None	

Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Click profile menu **2. Select "Logout"** **3. Confirm action**	**User: Staff** **Username: urs99**	**User is redirected to login page, session ends**	**As Expected,**	**Pass**

Project Name: AIUB Canteen Management System	**Test Designed by: Sha Sultan Sowhan**
Test Case ID: TC_13	**Test Designed date: 11/01/2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: Sha Sultan Sowhan**
Module Name: Discount Management	**Test Execution date: 20/01/2026**
Test Title: Verify that Admin can set discounts on products	
Description: Ensure Admin can create and apply discounts on products or combos.	
Precondition: Admin is logged in.	
Dependencies: Discount system is enabled.	

Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Log in to Dashboard 2. Select the product. 3. Click on "Set Discount".	**Username: admin, Password: xyz Product: Burger, Discount: 10%.**	**Discount is saved and reflected on product listing.**	**As expected,**	**Pass**

Project Name: AIUB Canteen Management System	**Test Designed by: Sha Sultan Sowhan**
Test Case ID: TC_14	**Test Designed date: 11/01/2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: Sha Sultan Sowhan**
Module Name: Login History	**Test Execution date: 20/01/2026**
Test Title: Verify that Admin can view login history of users	
Description: Ensure Admin can view all login activity logs.	
Precondition: Admin is logged in.	
Dependencies: Login logs are being recorded.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Dashboard 2. Click "View Login History"	Date Range: Last 7 days	The dashboard loads successfully. Login history table displays user activity.	As Expected,	Pass

Project Name: AIUB Canteen Management System	**Test Designed by: Sha Sultan Sowhan**
Test Case ID:TC_15	**Test Designed date: 11/01/2026**
Test Priority (Low, Medium, High): Medium	**Test Executed by: Sha Sultan Sowhan**
Module Name: Discount Management	**Test Execution date: 20/01/2026**
Test Title: Verify that Admin can set discounts on products	
Description: Ensure Admin can create and apply discounts on products or combos.	
Precondition: Admin is logged in.	
Dependencies: Discount system is enabled.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in to Dashboard 2. Select product. 3. Click on "Set Discount".	Username: admin, Password: xyz. Product: Burger, Discount: 10%.	Discount is saved and reflected on product listing.	As Expected,	Pass

6. CONCLUSION

AIUB Canteen Management System is designed to eradicate the problems and make the process of canteen management smooth and fast so that no time is wasted. The canteen makes extensive use of manpower for recording orders, managing inventory and generating bills currently. This over-reliance on labour-intensive processes can result in billing mistakes, delayed service periods, poor stock management and a loss of clarity around financial reporting. That's not just annoying for students, teachers and employees using the canteen daily, but it makes quality of service - something easily identified by all who can see problematic. canteen personnel and admin to perform with this visitor traffic with the university population increasing. We have made use of this methodology to systematically analyse the canteen`s requirements and together with the Admin, Staff, and Customers requirements. User stories, use cases diagrams, detailed functional descriptions have been utilized to drive the system design towards real needs. Estimates such as Lines of Code (LOC), COCOMO-based efforts, and effort schedule, project schedule and role identification also contribute to the organized view about what all resources, time and responsibilities are needed to ensure successful accomplishment of the project. The system also includes important features like secure login, inventory management, bill generation using pre-defined data, transaction reporting and customer feedback handling which collectively results in an all-in-one canteen solution. It worth mentioning that such an initiative does not only automate tasks but have its importance when in time to the effectiveness, correctness, and responsibility. The project replaces the manual process with a digital role-based system, paving the way for faster service, less human error and more reliable financial controls. Moreover, by wisely planning in this phase, the development will start with a very low risk and deliverables throughout the process. In Summary, project management phase complete with AIUB Canteen Management System is a feasible, useful and usable canteen solution to overcome the hurdles executing by present manual canteen. While no implementation of the software has been completed yet, by using this roadmap & design documentation we can be certain that all components needed are there. The efforts made till now will bring clarity for future developers, testers and stakeholders who can utilize a usable version for the system implementation which is designed to serve the real needs of the AIUB community.